Meridian

Audited volatility and tail-risk benchmark,
with a frozen forward test

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Holdout evaluation 2019-01-01 to 2026-06-29 · 2,168 trading dates × 11 assets · 20,906 out-of-sample forecasts per model

> Every number in this report is read programmatically from the saved results files (`results/benchmark_vol.json`, `results/benchmark_tail.json`, `results/forward/`) by `scripts/build_report_pdf.py`. Numbers produced before the harness audit are withdrawn and do not appear here.

1. Executive summary

Meridian is a state-space neural forecaster of daily volatility. This report covers a full audit of the benchmark harness, a re-run of the volatility benchmark against 15 standard models, a new tail-risk (VaR / Expected Shortfall) benchmark against 13 standard risk models, and the start of a frozen forward test. Model choices were made on 2012–2018 data only; all headline numbers are on the untouched 2019–2026 holdout.

Key results

- **Volatility: Meridian is the best model of 15.** Calibrated Meridian M0 reduces QLIKE loss by **-7.5%** [-11.3, -4.7] vs per-asset HAR-RV (DM p <0.001), wins on 11/11 assets, and is in the 90% Model Confidence Set. The best traditional model, LightGBM, achieves -5.3%.
- **Robust to the strongest baseline.** Against raw per-asset HAR (which scores better than its calibrated form), Meridian is -6.8% [-9.7, -4.5], p <0.001, and better in 14 of 15 calendar years.
- **Combination adds little.** A walk-forward combination of HAR + LightGBM + Meridian gives -8.2%, close to Meridian alone.
- **Tail risk: Meridian-based VaR/ES is the best of 13.** Meridian + EVT (selected on 2012–2018) lowers the joint VaR+ES loss vs GJR-GARCH + FHS (p <0.001) and the 1% VaR quantile loss by -9.2% [-12.7, -5.9], with ES calibrated to 0.98× realized tail losses.
- **But not significantly better than HAR-based tails.** Head-to-head vs HAR-RV + FHS: p 0.076 at 1% and 0.369 at 2.5%. Most of the tail gain comes from using range-based realized volatility, which HAR also uses.
- **Three proposed architecture improvements did not help** (per-asset HAR offset, asset embedding, early stopping) and were rejected on the development period — recorded as a negative result.
- **Forward test is live.** Models frozen on data through 2026-10-29; 13 pre-committed forecasts logged so far. No forward result yet.

> **Honest bottom line.** The volatility improvement over HAR is real and consistent. Its size against the strongest HAR form is -6.8% with a 95% interval of [-9.7, -4.5], so clearing the pre-registered 5% bar is likely but not certain. A better volatility forecast is not by itself a trading strategy, and no Sharpe ratio is claimed.

2. What was done

The work proceeded in five stages, each committed to the **harness-audit** branch.

Stage	Work	Outcome
1. Harness audit	Read-only review of the benchmark code; 8 defects found and fixed (next table).	All earlier numbers withdrawn (Amendment 6).
2. Volatility re-benchmark	15 models + Meridian variants, walk-forward, saved predictions, date-level statistics; variants chosen on 2012–2018.	Meridian M0 best; M1–M3 rejected on dev data.
3. Documentation clean-up	README, MODEL_CARD, ARCHITECTURE, WORLD_MODEL, RESEARCH, RESULTS: every pre-audit number removed or labelled.	Docs tables now generated from results files.
4. Tail-risk benchmark	13 VaR/ES models, FZ0 and quantile losses, Kupiec / Christoffersen / DQ backtests (Amendment 7).	Meridian + EVT best; ≈ HAR-filtered tails.
5. Forward test	Meridian M0 and per-asset HAR trained once, frozen and hashed; append-only log.	Live from the first trading day after freeze.

Harness defects found and fixed

Defect	Effect	Fix
Significance tests on ~40k stacked panel rows, HAC lag 0	p-values overstated by orders of magnitude	Diebold–Mariano on the date-level loss series, Newey–West lag; MCS and bootstrap over dates
Corrupt Yahoo FX price bars (36 stale, ~470 out-of-range)	Range-based RV read as ~0 (log ≈ −27.6), dominating QLIKE	Drop stale bars, clip open/close into [low, high]
Pooled HAR baseline (one regression across equities and FX)	Handicapped reference model	Per-asset HAR as the reference
5- and 22-day labels crossing into the test block	Label leakage at longer horizons	Purge training rows by label-end date
TimeMixer scaled with full-sample statistics	Look-ahead	Train-only scaling per fold
EWMA / GARCH forecasts one day stale	Unfairly weak baselines	True one-step-ahead forecasts
Raw and calibrated tables on different rows	Sample change mixed with model change	Every table scored on one common row support
No saved predictions; hand-copied README numbers	Nothing re-checkable	Predictions saved; scoring and docs regenerated from files

Automated tests (11, all passing) check: no training label crosses a test start, calibration has no look-ahead, the DM test is calibrated under the null, QLIKE / FZ0 / quantile losses are minimized at the truth, VaR backtests are calibrated under the null, and tail forecasts never use future returns.

3. Data and evaluation protocol

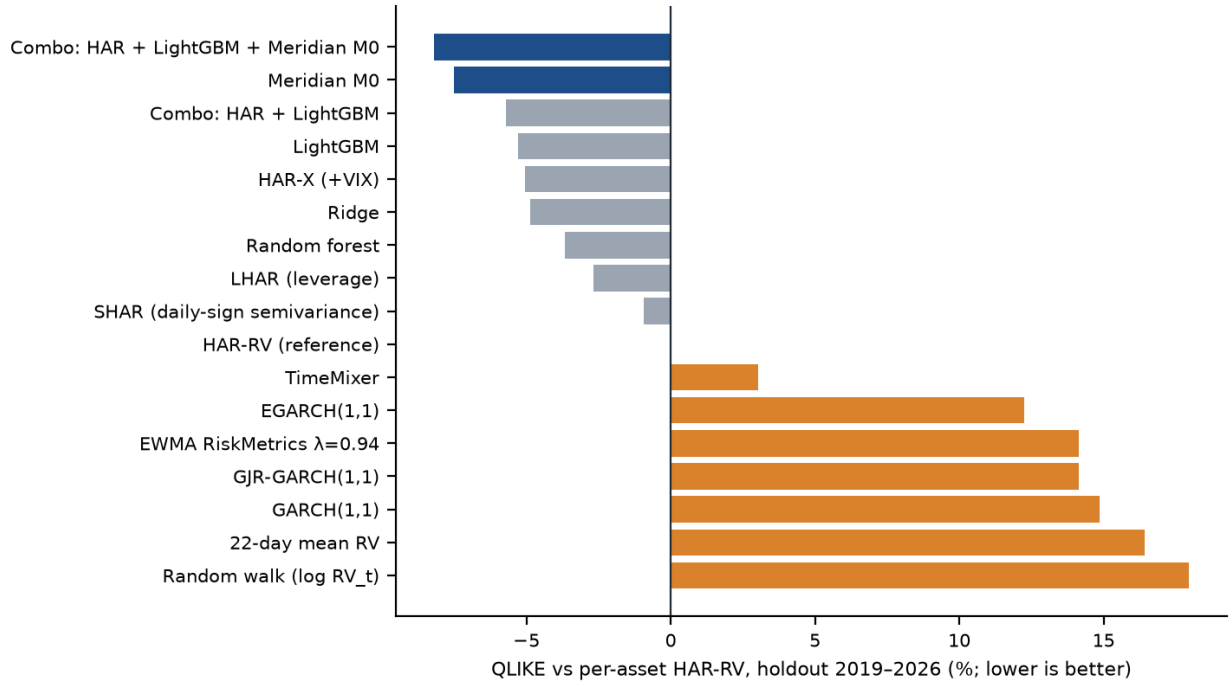
- **Data:** daily OHLC from Yahoo for 8 US equities/ETFs (SPY, QQQ, IWM, AAPL, MSFT, JPM, XOM, JNJ) and 3 FX pairs (EURUSD, USDJPY, GBPUSD); VIX and Treasury yields from FRED; 2007–2026. TLT and GLD are scored separately as assets the pooled models never trained on.
- **Target:** next-day log realized variance, Garman–Klass range estimator (no intraday data). Tail benchmark target: next-day close-to-close log return.
- **Walk-forward:** expanding window, first test block after ~5 years, 378-day test blocks (12 folds), 22-day embargo, training rows purged by label-end date; all scalers fit on training rows only.
- **Calibration:** every model gets the same walk-forward affine recalibration (past data only), so comparisons are like for like.
- **Selection rule:** development period = test dates before 2019-01-01, used for every choice; holdout = 2019-01-01 onward, used only for the final report. Declared in PREREGISTRATION.md before results were seen.
- **Statistics:** QLIKE loss (primary), Diebold–Mariano test on the date-level loss differential with Newey–West HAC, moving-block bootstrap 95% CIs (block 22), Model Confidence Set at 90%, Mincer–Zarnowitz R^2, per-asset time-series rank IC, directional hit rate.
- **Meridian M0:** SSM belief core with JEPA predictor, QLIKE-trained log-variance head, initialized at the HAR solution, 5-seed ensemble, retrained every fold.

4. Volatility forecasting results

Holdout 2019-01-01 to 2026-06-29, all models calibrated, reference = per-asset HAR-RV. Green rows are Meridian; red rows are worse than HAR. Lower QLIKE is better.

Model	Family	QLIKE	vs HAR (95% CI)	DM p	MZ R^2	IC	Hit rate	Assets beating HAR	In 90% MCS
GR combo: HAR + LightGBM + Meridian M0 (pooled HAR anchor)	Combo	0.3022	-8.2% [-12.4, -5.2]	<0.001	0.413	0.594	68.3%	11/11	yes
Meridian M0 (pooled HAR anchor)	Meridian	0.3045	-7.5% [-11.3, -4.7]	<0.001	0.410	0.591	68.1%	11/11	no
GR combo: HAR + LightGBM	Combo	0.3105	-5.7% [-9.3, -3.1]	<0.001	0.400	0.584	67.8%	10/11	no
LightGBM	ML	0.3118	-5.3% [-9.3, -2.5]	<0.001	0.399	0.583	67.9%	10/11	no
HAR-X (+VIX)	HAR	0.3126	-5.0% [-7.7, -2.9]	<0.001	0.391	0.578	66.9%	10/11	no
Ridge	ML	0.3132	-4.9% [-8.7, -2.1]	<0.001	0.397	0.578	67.2%	9/11	no
Random forest	ML	0.3171	-3.7% [-6.8, -1.4]	<0.001	0.388	0.573	67.6%	11/11	no
LHAR (leverage)	HAR	0.3204	-2.7% [-5.4, -0.5]	0.011	0.386	0.569	68.0%	10/11	no
SHAR (daily-sign semivariance)	HAR	0.3261	-0.9% [-1.6, -0.4]	0.004	0.372	0.566	67.2%	8/11	no
HAR-RV, pooled	HAR	0.3291	-0.0% [-0.4, +0.3]	0.285	0.368	0.563	66.8%	4/11	no
HAR-RV, per asset (reference)	HAR	0.3292	+0.0%	—	0.367	0.562	66.9%	0/11	no
TimeMixer	Deep	0.3392	+3.0% [+0.3, +6.1]	0.957	0.345	0.542	67.0%	3/11	no
EGARCH(1,1)	GARCH	0.3695	+12.2% [+8.1, +16.8]	1.000	0.321	0.515	68.2%	0/11	no
EWMA RiskMetrics λ=0.94	Naive	0.3758	+14.1% [+7.9, +21.8]	1.000	0.298	0.513	66.8%	0/11	no
GJR-GARCH(1,1)	GARCH	0.3758	+14.1% [+9.9, +18.9]	1.000	0.315	0.504	68.1%	0/11	no
GARCH(1,1)	GARCH	0.3782	+14.9% [+11.0, +18.6]	1.000	0.293	0.490	67.7%	0/11	no
22-day mean RV	Naive	0.3833	+16.4% [+9.1, +25.9]	1.000	0.283	0.501	67.0%	0/11	no
Random walk (log RV_t)	Naive	0.3884	+18.0% [+12.9, +24.7]	1.000	0.266	0.472	62.3%	0/11	no

Hit rate (directional accuracy of the up/down call) is shown but does not separate the models: nearly all score 66–68% because volatility persistence makes the direction easy to call.

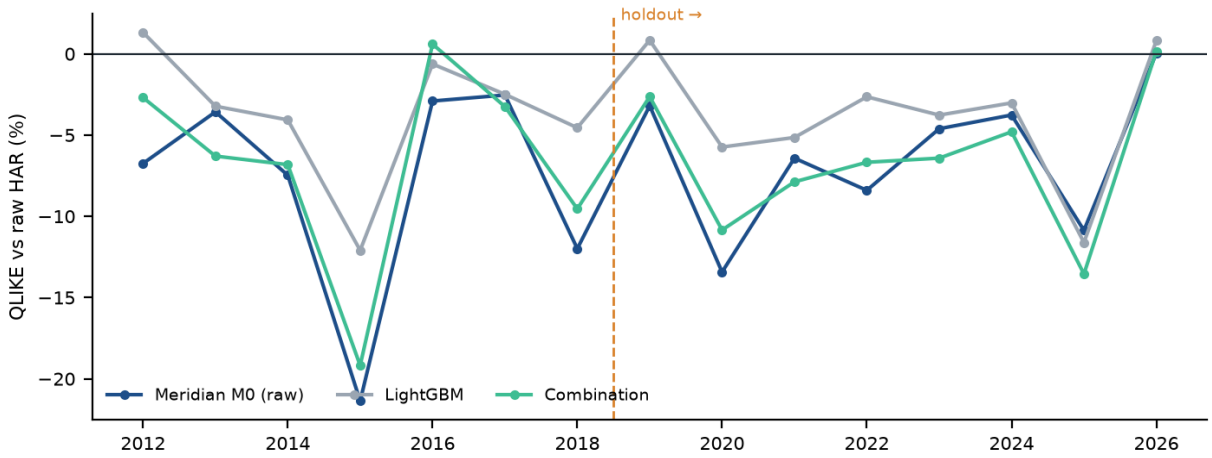


4.1 Conservative check against the strongest HAR

Raw per-asset HAR scores better than its calibrated version on this holdout, so the most demanding comparison is against raw HAR.

Model	QLIKE	vs raw per-asset HAR (95% CI)	DM p	Assets beating
GR combo: HAR + LightGBM + Meridian M0 (pooled HAR anchor)	0.3022	-7.0% [-9.5, -4.9]	<0.001	11/11
Meridian M0 (pooled HAR anchor) — raw	0.3029	-6.8% [-9.7, -4.5]	<0.001	11/11
Meridian M0 (pooled HAR anchor) — calibrated	0.3045	-6.3% [-8.6, -4.3]	<0.001	11/11
GR combo: HAR + LightGBM	0.3105	-4.4% [-6.6, -2.7]	<0.001	10/11
LightGBM — calibrated	0.3118	-4.0% [-6.5, -2.3]	<0.001	10/11
HAR-RV, per asset — raw (reference)	0.3249	+0.0%	—	0/11
HAR-RV, per asset — calibrated	0.3292	+1.3% [-0.2, +3.9]	0.914	0/11

4.2 Consistency over time



Meridian beats raw HAR in 14 of 15 calendar years (2026 is a partial year).

4.3 Per asset (holdout, vs raw per-asset HAR)

Asset	Meridian M0 (raw)	Meridian M0 (calibrated)	LightGBM (calibrated)	Combination
AAPL	-7.3%	-7.1%	-2.8%	-6.9%
EURUSD	-4.2%	-3.3%	-1.5%	-3.0%
GBPUSD	-4.1%	-1.3%	-4.1%	-4.1%
IWM	-11.0%	-10.6%	-7.9%	-10.1%
JNJ	-3.7%	-3.9%	-1.7%	-4.2%
JPM	-9.1%	-9.0%	-9.9%	-11.4%
MSFT	-8.5%	-8.4%	-5.0%	-8.3%
QQQ	-12.6%	-12.7%	-8.8%	-13.7%
SPY	-10.3%	-10.4%	-10.1%	-12.5%
USDJPY	-1.1%	-1.0%	+3.2%	-1.0%
XOM	-7.9%	-6.6%	-1.1%	-6.8%

4.4 By market regime (full out-of-sample period, VIX terciles, vs calibrated HAR)

Model	Calm VIX tercile	Normal	Stress
Meridian M0 (pooled HAR anchor)	-4.1% (p <0.001)	-5.2% (p <0.001)	-14.1% (p 0.009)
GR combo: HAR + LightGBM + Meridian M0 (pooled HAR anchor)	-4.9% (p <0.001)	-6.4% (p 0.034)	-15.3% (p 0.005)
LightGBM	-2.9% (p <0.001)	-4.1% (p 0.203)	-10.9% (p 0.004)
HAR-X (+VIX)	-1.8% (p <0.001)	-2.8% (p <0.001)	-9.6% (p 0.004)
LHAR (leverage)	-1.2% (p <0.001)	-2.2% (p <0.001)	-7.1% (p 0.035)
GJR-GARCH(1,1)	+9.6% (p 1.000)	+8.1% (p 1.000)	+12.9% (p 0.984)
EWMA RiskMetrics λ=0.94	+6.2% (p 1.000)	+9.0% (p 1.000)	+26.9% (p 0.999)

4.5 Longer horizons (holdout, calibrated, Meridian auxiliary heads vs HAR)

Horizon	HAR QLIKE	Meridian QLIKE	Meridian vs HAR (95% CI)	DM p	Assets beating
5-day	0.1900	0.1718	-9.6% [-14.5, -4.9]	<0.001	11/11
22-day	0.2086	0.2067	-0.9% [-4.5, +6.7]	0.356	7/11

At 5 and 22 days only HAR and Meridian have horizon-specific forecasts in this run; the full 15-model comparison exists for the 1-day horizon.

4.6 Assets never seen in pooled training (TLT, GLD)

Model (TLT + GLD, never in pooled training)	QLIKE	vs HAR (95% CI)	DM p
Meridian M0 (pooled HAR anchor)	0.3086	-7.1% [-12.4, -2.1]	0.002
HAR-X (+VIX)	0.3265	-1.7% [-6.6, +1.9]	0.181
HAR-RV, per asset (reference)	0.3320	+0.0%	—
LightGBM	0.3473	+4.6% [-0.4, +10.4]	0.961
EWMA RiskMetrics λ=0.94	0.3594	+8.2% [+4.2, +12.8]	1.000
GJR-GARCH(1,1)	0.3861	+16.3% [+6.4, +29.5]	0.999

4.7 Proposed improvements — development-period ablation

Three architecture changes were tried (3 seeds each) and compared only on 2012–2018. None beat the existing configuration, so M0 was kept. The combination was selected but is practically tied with Meridian alone.

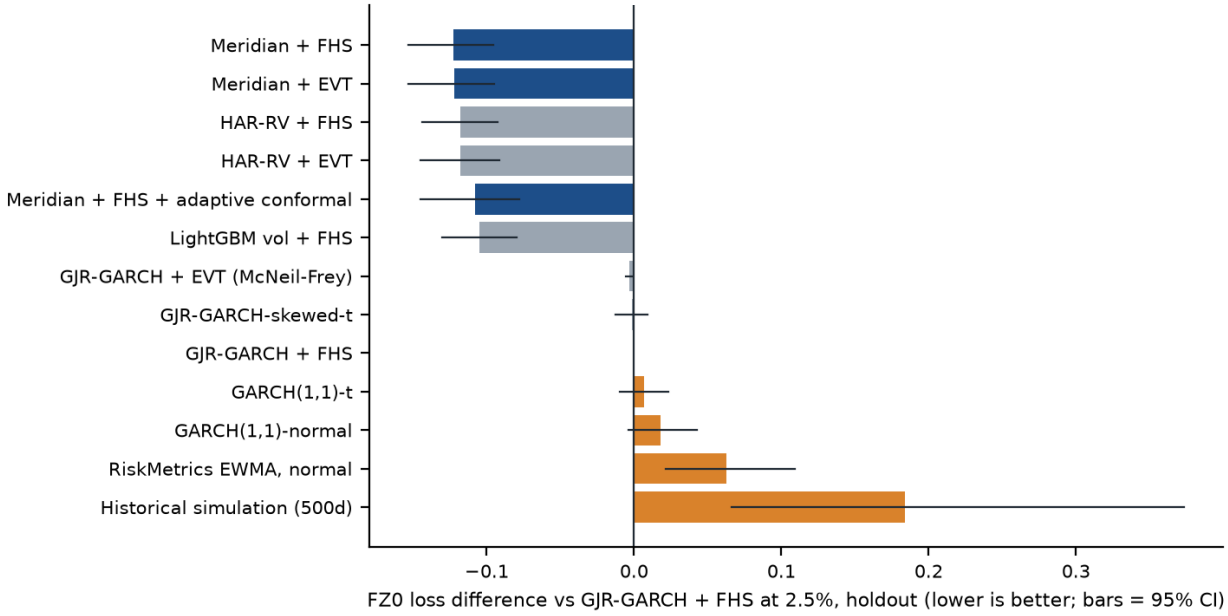
Candidate	What it is	Dev QLIKE (2012–2018)	vs HAR
GR combo: HAR + LightGBM + Meridian M0 (pooled HAR anchor)	existing: pooled HAR-anchored init	0.3353	-10.4%
GR combo: HAR + LightGBM + Meridian M2 (M1 + asset embedding)	new: M1 + asset embedding	0.3354	-10.3%
GR combo: HAR + LightGBM + Meridian M1 (per-asset HAR offset)	new: per-asset HAR offset	0.3356	-10.3%
GR combo: HAR + LightGBM + Meridian M3 (M1 + early stopping)	new: M1 + early stopping	0.3381	-9.6%
Meridian M0 (pooled HAR anchor)	existing: pooled HAR-anchored init	0.3404	-9.0%
Meridian M2 (M1 + asset embedding)	new: M1 + asset embedding	0.3430	-8.3%
Meridian M1 (per-asset HAR offset)	new: per-asset HAR offset	0.3432	-8.3%
Meridian M3 (M1 + early stopping)	new: M1 + early stopping	0.3477	-7.1%
HAR-RV, per asset (reference)	reference	0.3741	+0.0%

5. Tail-risk results (VaR and Expected Shortfall)

Next-day VaR and ES of close-to-close returns at 1%, 2.5% and 5%; holdout 2019-01-01 to 2026-06-29, 2,168 dates × 11 assets. Reference: GJR-GARCH with filtered historical simulation. Volatility-based tails filter returns through a saved out-of-sample volatility forecast and read the tail from past standardized returns (empirically, or with an EVT/GPD fit). Primary score: FZ0 loss, which scores VaR and ES jointly. CC = Christoffersen conditional coverage, DQ = Engle–Manganelli dynamic quantile test; an asset passes at p > 0.05. ES ratio ≈ 1 means the forecast ES matched realized tail losses.

5.1 Joint VaR + ES at 2.5%

Model	FZ0 loss	Δ vs reference (95% CI)	DM p	In MCS	Violations (2.5% target)	Assets pass CC · DQ	ES ratio
Meridian + FHS	-3.6394	-0.1226 [-0.1530, -0.0959]	<0.001	yes	2.54%	11/11 · 10/11	0.97
Meridian + EVT	-3.6387	-0.1218 [-0.1533, -0.0949]	<0.001	yes	2.49%	11/11 · 10/11	0.98
HAR-RV + FHS	-3.6289	-0.1182 [-0.1440, -0.0927]	<0.001	yes	2.58%	11/11 · 8/11	0.99
HAR-RV + EVT	-3.6284	-0.1179 [-0.1449, -0.0912]	<0.001	yes	2.57%	11/11 · 7/11	0.99
Meridian + FHS + adaptive conformal	-3.6257	-0.1080 [-0.1448, -0.0782]	<0.001	no	2.47%	11/11 · 11/11	0.98
LightGBM vol + FHS	-3.6208	-0.1048 [-0.1306, -0.0796]	<0.001	no	2.58%	11/11 · 10/11	0.98
GJR-GARCH-skewed-t	-3.5294	-0.0013 [-0.0124, +0.0093]	0.404	no	2.60%	10/11 · 10/11	0.99
GJR-GARCH + EVT (McNeil-Frey)	-3.5290	-0.0030 [-0.0054, -0.0007]	0.007	no	2.67%	11/11 · 11/11	0.98
GJR-GARCH + FHS (reference)	-3.5255	—	—	no	2.71%	11/11 · 11/11	0.98
GARCH(1,1)-t	-3.5203	+0.0067 [-0.0097, +0.0234]	0.799	no	2.87%	10/11 · 7/11	1.00
GARCH(1,1)-normal	-3.5049	+0.0179 [-0.0036, +0.0427]	0.951	no	3.01%	9/11 · 7/11	1.14
RiskMetrics EWMA, normal	-3.4538	+0.0626 [+0.0215, +0.1094]	0.999	no	3.48%	4/11 · 1/11	1.16
Historical simulation (500d)	-3.3320	+0.1836 [+0.0660, +0.3735]	0.997	no	2.80%	5/11 · 1/11	1.07



5.2 Head-to-head: Meridian + EVT vs HAR-RV + FHS

Both sides use a realized-volatility forecast, so this isolates Meridian's volatility forecast from HAR's. The differences favor Meridian but are not statistically significant.

Level	FZ0 Δ (95% CI)	FZ0 DM p	Quantile loss vs HAR-FHS (95% CI)	Quantile DM p
1.0%	-0.0100 [-0.0461, +0.0168]	0.245	-3.1% [-7.4, +0.2]	0.076
2.5%	-0.0037 [-0.0333, +0.0175]	0.369	-2.0% [-6.2, +1.1]	0.159
5.0%	-0.0013 [-0.0246, +0.0144]	0.439	-1.1% [-4.1, +1.0]	0.203

5.3 VaR at 1% (quantile loss on returns scaled per asset)

Model	Quantile loss ×10³	vs reference (95% CI)	DM p	Violations	Assets pass Kupiec · CC · DQ (of 11)
Meridian + EVT	22.27	-9.2% [-12.7, -5.9]	<0.001	0.96%	11 · 11 · 10
Meridian + FHS	22.32	-9.0% [-12.5, -5.7]	<0.001	1.04%	11 · 11 · 10
LightGBM vol + FHS	22.59	-7.9% [-10.8, -4.7]	<0.001	1.11%	11 · 11 · 10
HAR-RV + EVT	22.98	-6.3% [-9.7, -2.5]	<0.001	0.99%	11 · 10 · 7
HAR-RV + FHS	22.99	-6.3% [-9.6, -2.7]	<0.001	1.09%	11 · 10 · 6
Meridian + FHS + adaptive conformal	23.26	-5.2% [-9.1, -1.5]	<0.001	1.00%	11 · 11 · 11
CAViaR asymmetric slope	24.21	-1.3% [-2.9, +0.6]	0.181	1.11%	10 · 10 · 10
GARCH(1,1)-t	24.39	-0.6% [-2.7, +1.7]	0.357	1.19%	9 · 11 · 8
GJR-GARCH-skewed-t	24.47	-0.3% [-1.4, +1.1]	0.437	1.01%	11 · 11 · 11
GJR-GARCH + EVT (McNeil-Frey)	24.51	-0.1% [-0.5, +0.4]	0.411	1.05%	11 · 11 · 11
GJR-GARCH + FHS (reference)	24.53	+0.0%	—	1.12%	11 · 11 · 10
GARCH(1,1)-normal	25.08	+2.2% [-1.9, +7.3]	0.828	1.80%	2 · 2 · 2
RiskMetrics EWMA, normal	26.37	+7.5% [+0.1, +17.1]	0.972	2.10%	0 · 0 · 0
Historical simulation (500d)	30.97	+26.3% [+6.7, +57.4]	0.980	1.43%	8 · 4 · 0

CAViaR forecasts VaR only, so it appears in the VaR tables but not in the joint VaR+ES table.

5.4 VaR at 5%

Model	Quantile loss ×10³	vs reference (95% CI)	DM p	Violations	Assets pass Kupiec · CC · DQ (of 11)
Meridian + EVT	77.07	-5.1% [-6.7, -3.6]	<0.001	4.95%	11 · 11 · 9
Meridian + FHS	77.17	-5.0% [-6.5, -3.5]	<0.001	4.97%	11 · 11 · 9
Meridian + FHS + adaptive conformal	77.74	-4.3% [-6.2, -2.5]	<0.001	4.90%	11 · 11 · 9
LightGBM vol + FHS	77.78	-4.2% [-5.6, -2.9]	<0.001	5.11%	11 · 11 · 9
HAR-RV + FHS	77.94	-4.0% [-5.7, -2.1]	<0.001	5.13%	11 · 11 · 8
HAR-RV + EVT	77.95	-4.0% [-5.7, -2.1]	<0.001	5.12%	11 · 11 · 7
GJR-GARCH-skewed-t	80.79	-0.5% [-2.0, +0.6]	0.222	4.89%	9 · 10 · 10
GJR-GARCH + EVT (McNeil-Frey)	81.12	-0.1% [-0.2, -0.0]	0.063	5.25%	11 · 10 · 10
CAViaR asymmetric slope	81.21	-0.0% [-1.2, +1.0]	0.788	4.95%	8 · 8 · 9
GJR-GARCH + FHS (reference)	81.22	+0.0%	—	5.37%	11 · 10 · 9
GARCH(1,1)-normal	81.57	+0.4% [-0.5, +1.4]	0.925	4.99%	8 · 10 · 9
GARCH(1,1)-t	81.83	+0.7% [-0.0, +1.6]	0.969	5.37%	8 · 10 · 7
RiskMetrics EWMA, normal	83.19	+2.4% [+0.8, +4.0]	0.999	5.48%	10 · 10 · 3
Historical simulation (500d)	89.55	+10.3% [+3.4, +20.1]	0.996	5.17%	11 · 5 · 1

5.5 Development period, 2.5% (used to select the Meridian tail variant)

Model	FZ0 loss	Δ vs reference (95% CI)	DM p	In MCS	Violations (2.5% target)	Assets pass CC · DQ	ES ratio
HAR-RV + FHS	-3.7832	-0.0998 [-0.1247, -0.0724]	<0.001	yes	2.61%	10/11 · 7/11	1.04
HAR-RV + EVT	-3.7830	-0.0981 [-0.1230, -0.0710]	<0.001	yes	2.46%	9/11 · 8/11	1.04
Meridian + EVT	-3.7753	-0.0865 [-0.1159, -0.0585]	<0.001	yes	2.47%	10/11 · 9/11	1.04
Meridian + FHS	-3.7746	-0.0862 [-0.1164, -0.0573]	<0.001	yes	2.63%	10/11 · 8/11	1.03
Meridian + FHS + adaptive conformal	-3.7547	-0.0677 [-0.0976, -0.0393]	<0.001	no	2.53%	10/11 · 9/11	1.04
LightGBM vol + FHS	-3.7514	-0.0595 [-0.0850, -0.0327]	<0.001	no	2.68%	11/11 · 9/11	1.03
GJR-GARCH + EVT (McNeil-Frey)	-3.7034	-0.0002 [-0.0027, +0.0024]	0.452	no	2.23%	9/11 · 9/11	1.05
GJR-GARCH + FHS (reference)	-3.7016	—	—	no	2.33%	9/11 · 9/11	1.03
GJR-GARCH-skewed-t	-3.6911	+0.0161 [-0.0003, +0.0293]	0.983	no	1.88%	5/11 · 8/11	1.05
GARCH(1,1)-t	-3.6842	+0.0204 [+0.0087, +0.0329]	0.999	no	2.21%	8/11 · 10/11	1.06
GARCH(1,1)-normal	-3.6804	+0.0187 [+0.0036, +0.0379]	0.980	no	2.33%	10/11 · 9/11	1.18
RiskMetrics EWMA, normal	-3.6207	+0.0676 [+0.0230, +0.1249]	0.995	no	3.26%	5/11 · 3/11	1.21
Historical simulation (500d)	-3.5578	+0.1409 [+0.0936, +0.1979]	1.000	no	2.84%	4/11 · 0/11	1.04

On 2012–2018 the HAR-based tails were slightly ahead of Meridian's: the ranking at the very top is not stable across periods.

6. Forward test

A backtest can be tuned, consciously or not, to the data it is scored on. The forward test removes that risk: the configuration is frozen and then scored only on days that had not happened when each forecast was logged.

Item	Value
Frozen at (UTC)	2026-11-01T16:19:03+00:00
Training data through	2026-10-29
Model	Meridian M0, 5 seeds, pooled HAR-anchored init
Reference	per-asset HAR-RV
Training rows	54,481
Code commit at freeze	108f349
Integrity	SHA-256 of every weight file checked before each forecast
Forecasts logged	13 (13 pre-committed; 11 on the 11 core assets)
First forecast origin	2026-10-30
Scored results	none yet — needs months of new data for a meaningful read

Daily operation: `python scripts/forward_test.py predict` after each close; `python scripts/forward_test.py score` to score. A forecast counts only if its origin was the latest available bar when it was logged. The forward test keeps its own data snapshot so the audited benchmark inputs are never overwritten.

7. Limitations and what is not claimed

- Daily range-based realized variance only. Without intraday data, HARQ, HAR-CJ jump models, Realized GARCH and a true semivariance SHAR could not be tested; the SHAR here is a daily-sign approximation.
- Free, delayed Yahoo / FRED data; 11 core assets. Results may not transfer to other asset classes.
- A better volatility or VaR forecast is not alpha: no trading, Sharpe or P&L figure is claimed.
- The Diebold–Mariano test omits the Harvey–Leybourne–Newbold small-sample correction; with ~2,200 test dates the effect is negligible.
- Regime, factor-state, network, combiner, continual-learning and portfolio modules exist in code but have not been re-validated; their earlier numbers are withdrawn.

8. Remaining work

Priority	Item	Why
1	Run the forward test daily for several months	The only evidence free of backtest selection
2	Re-validate the other modules (regime, factor, combiner, …)	No audited numbers exist for them
3	Full 15-model comparison at 5- and 22-day horizons	Currently Meridian vs HAR only
4	Intraday (5-minute) data	Enables HARQ / jump models and a cleaner target; may narrow the CI
5	More assets (bonds, commodities, futures)	Tighter confidence intervals, broader validity

Appendix — reproducibility

All results regenerate from the repository. Scoring reads only saved predictions and reproduces the results files byte for byte.

Command	Produces
python -m pytest tests -q	11 harness tests
BENCH_VARIANTS=M0 BENCH_ENSEMBLE=5 python scripts/benchmark_vol.py	results/benchmark_vol_predictions.parquet (~20 min)
python scripts/score_vol_benchmark.py	results/benchmark_vol.json / .csv
python scripts/render_benchmark_md.py	results/benchmark_vol.md + README / MODEL_CARD tables
python scripts/benchmark_tail.py	results/benchmark_tail_predictions.parquet (~2 min)
python scripts/score_tail_benchmark.py	results/benchmark_tail.json / .md + doc tables
python scripts/forward_test.py predict | score	results/forward/forecast_log.csv, forward_score.json
python scripts/build_report_pdf.py	this report

Pre-registration

PREREGISTRATION.md Amendment 6 (harness audit, volatility selection rule and selection record) and Amendment 7 (tail-risk benchmark, forward test and selection record) were written before the corresponding holdout results were computed.

Model glossary

Term	Meaning
QLIKE	Standard loss for variance forecasts; robust to noise in the realized-variance proxy
HAR-RV	Heterogeneous autoregression on daily, weekly and monthly realized variance (Corsi 2009)
GJR-GARCH / EGARCH	GARCH variants allowing a larger response to negative returns
FHS	Filtered historical simulation: tail read from past returns divided by the volatility forecast
EVT	Extreme-value theory: a Generalized Pareto fit to the tail of standardized returns
CAViaR	Conditional autoregressive VaR estimated by quantile regression (Engle–Manganelli 2004)
FZ0 loss	Joint VaR + ES scoring function (Fissler–Ziegel; Patton, Ziegel and Chen 2019)
DM test	Diebold–Mariano test of equal predictive accuracy
MCS	Model Confidence Set: the models not statistically worse than the best